

Problem Set 8

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1 More probability theory!

For each problem, it is not enough to simply state the answer, please explain how you came to your answer.

Problem 1: A biased coin comes up heads 60% of the time. The coin is tossed 200 times. Let X be the random variable representing the number of heads in these 200 coin flips. Use Chebyshev's inequality to lower bound the probability that X is strictly between 100 and 140.

Problem 2: Suppose we have a non-negative random variable with expectation value 50. Then:

1. Derive an upper bound for the probability that X is at least 100. (Note: Skip ONLY this part of the problem as you had it on last week's homework)
2. Suppose we know that the variance of X is 25. Then can we obtain a better bound for part (1) of this problem?

Problem 3: Suppose I have two bins, which I'll call B_1 and B_2 . Let's say B_1 contains 2 red balls and 8 blue balls, while B_2 contains 7 red balls and 3 blue balls. Suppose Bob randomly chooses one of the two bins B_1 and B_2 with equal probability, $1/2$ of choosing each. Now suppose Bob then randomly picks one ball out of the bin he has chosen, without ever telling us which bin that is. Given that the ball Bob picked is red, what is the probability that Bob chose bin B_1 ?